

PHASE 1 PROJECT DIARY AND TECHNICAL REPORT

*Sensor-Based Machine Learning and Deep Learning Pipeline
Using the IDNet Sample Dataset*

Prepared by
Sai Sreevatsal
M. Rahul
M.S. Karthik

Purpose: Detailed record of Phase 1 procedures, experiments, validation, results, observations, and current status.

Dataset note: IDNet is used here strictly as a sample dataset for pipeline validation; it is not being treated as Parkinson disease data.

1. Executive Summary

The first phase of the project was designed as a controlled sample-data experiment. The IDNet dataset was used to validate the complete pipeline before moving to the actual Parkinson-labelled dataset. The work progressed from raw inertial sensor logs to data-quality checks, synchronization, fixed-length windowing, handcrafted feature extraction, traditional machine-learning models, deep-learning models, normalization experiments, multiple validation strategies, FAR/FRR analysis, confusion matrices, and a simple model-integration experiment.

Current status

The complete processing and evaluation framework is operational. The main validation experiments and model-integration experiment are complete. The remaining Phase 1 gap is clarification of two guide-specific normalization items: the exact intended definition of separate “Standard-deviation normalization” and the procedure meant by “Canonical Multi-Vendor Schema Mapping.”

Key outcomes at a glance

Area	Result
Sample dataset structure	50 users; 135 recording folders
Main sensor channels	Accelerometer (3), Gyroscope (3), Magnetometer (3) = 9 channels
Fixed DL window	5 seconds at 60 Hz = 300 samples x 9 channels
Handcrafted feature set	90 features per window
RF final baseline test accuracy	74.18%
SVM final baseline test accuracy	74.79%
1D CNN final test accuracy	74.69%
CNN-LSTM final test accuracy	53.35%
Best completed normalization/validation result	Stratified 5-Fold + Linear (Min-Max) + SVM: 89.45% pooled accuracy, 2.73% FAR, 19.34% FRR
Common-holdout ensemble	RF + SVM + CNN majority voting: 84.80% accuracy, 3.80% FAR, 25.03% FRR

2. Phase 1 Requirement Given by the Guide

The guide specified the following experimental structure:

- Apply different normalization techniques before model training.
- Evaluate each technique under four validation methods: K-Fold, Stratified K-Fold, Holdout, and Time-Series Split.
- Calculate Accuracy, FAR, and FRR for each combination.
- Include confusion-matrix plots for the best-performing and worst-performing combinations.
- Prepare a structured comparison and a brief conclusion identifying the combination with the highest accuracy and lowest FAR/FRR.
- Normalize and validate sensor data consistently so that preprocessing choices do not leak information from validation/test data.

2.1 Normalization techniques named by the guide

#	Guide-listed method	Status in current work	Note
1	Decimal-place normalization	Implemented	Per-feature scaling based on training-data magnitude.
2	Z-score normalization	Implemented	StandardScaler; training statistics only.
3	Dynamic Z-score normalization	Implemented operationally	Implemented as row/sample-wise standardization; exact guide definition still to be confirmed.
4	Linear normalization	Implemented	Implemented as Min-Max scaling

#	Guide-listed method	Status in current work	Note
			to [0,1].
5	Non-linear power law normalization	Implemented operationally	Signed power transform with $\gamma=0.5$; exact guide definition still to be confirmed.
6	Standard-deviation normalization	Not run separately	Overlaps conceptually with Z-score without a separate guide formula.
7	Adaptive sequential normalization	Implemented operationally	Running mean/variance approach; exact guide definition still to be confirmed.
8	Canonical Multi-Vendor Schema Mapping	Not implemented	Not a standard numeric scaling formula; guide-specific procedure needed.

3. Dataset and Data Understanding

The IDNet directory contains recording folders named in the form uXXX_wYYY, where the subject/user identifier is followed by a recording/session identifier. The sample experiment inspected 50 users (u001-u050) and 135 recording folders.

3.1 Files observed within recordings

- accelerometer.log
- gyroscope.log
- magnetometer.log
- linearaccelerometer.log
- gravity.log
- rotvec.log

The modeling pipeline selected the three primary inertial sensors: accelerometer, gyroscope, and magnetometer. Each contributes X, Y, Z channels, giving 9 channels per time point.

4. Data Quality Checks

Before modeling, the recordings were checked for file existence, missing values, duplicate timestamps, timestamp ordering, duration consistency, sampling-rate behavior, constant signals, and duration differences between sensors.

Quality outcome	Count/observation
Total recording folders inspected	135
Technically valid recordings	119
Review-required recordings	16
Typical reason for review	Timestamp-order or related consistency issue

4.1 Detailed timestamp investigation: u033_w006

u033_w006 was marked REVIEW REQUIRED. It had no missing samples and no duplicate timestamps, but timestamp order contained negative jumps across all three primary sensors.

Sensor	Negative jumps
Accelerometer	-1.123944907 s and -0.176821636 s
Gyroscope	-1.127912192 s and -0.180483745 s
Magnetometer	-1.116712241 s and -0.165682720 s

Because the anomalies were systematic across sensors, the recording was preserved as REVIEW REQUIRED rather than silently deleted. This became part of the project data-quality record.

5. Sampling-Rate Analysis and Synchronization

The three sensors do not originally sample at the same rate. A representative example showed approximately 119 Hz for accelerometer, approximately 199 Hz for gyroscope, and approximately 59.5 Hz for magnetometer. Direct row-by-row merging would therefore be invalid.

5.1 Synchronization procedure

1. Load the three sensor logs.
2. Standardize their timestamp and X/Y/Z column names.
3. Find the common start time as the latest starting timestamp among the three sensors.
4. Find the common end time as the earliest ending timestamp.
5. Create a fixed master timeline at 60 Hz.
6. Interpolate X/Y/Z values of each sensor onto the common timeline.
7. Build a synchronized 9-channel table containing accelerometer, gyroscope, and magnetometer values plus elapsed time.

Why 60 Hz?

The magnetometer stream was already close to 60 Hz, so a 60 Hz common timeline provided a practical base rate for the fused signal. The exact target used in the implemented processing function was 60 Hz.

6. Windowing

Continuous synchronized data were divided into fixed-size windows for both handcrafted-feature models and deep-learning models.

Parameter	Value
Window duration	5 seconds
Target sampling rate	60 Hz
Samples per window	300
Sensor channels	9
Overlap	50%
Deep-learning input shape	300 x 9 per window; transposed to 9 x 300 for Conv1D

An earlier version produced variable window lengths such as 297 samples because different recordings were being treated with different effective rates. The pipeline was corrected by using the fixed 60 Hz master timeline. This made every full window exactly 300 x 9.

7. Handcrafted Feature Extraction

Each 300 x 9 window was summarized into 90 numerical features for traditional machine learning. Ten features were computed for each of the 9 sensor channels.

Feature family	Features used
Time domain	Mean, standard deviation, minimum, maximum, median, range, RMS, zero-crossing rate
Frequency domain	Dominant frequency, spectral energy
Total	10 features per channel x 9 channels = 90 features

The frequency-domain features were calculated from the real FFT after removing the DC component. A nominal sampling rate of 59.47 Hz was used in the feature extraction function for frequency calculation.

8. Baseline Model Experiments

The initial classification task used five selected user labels: u018, u023, u024, u033, and u036. This is a USER-IDENTIFICATION experiment on sample data, not a Parkinson disease classification experiment.

8.1 Random Forest

Random Forest was selected as a strong tabular-data baseline. It can model nonlinear feature interactions, works naturally with numerical features, and provides feature-importance information.

RF experiment	Accuracy	Confusion-matrix observation
Initial 63-feature experiment	67.97%	Strong confusion involving u033; baseline was improved after feature expansion.
Improved 90-feature experiment	74.18%	u024 improved substantially; u033 remained difficult.

Important feature-importance outputs included gyro_y_dominant_freq, gyro_z_dominant_freq, gyro_x_dominant_freq, gyro_z_zcr, acc_y_spectral_energy, acc_y_zcr, and other frequency-domain or RMS features.

8.2 Random Forest confusion matrices

True \ Pred.	u018	u023	u024	u033	u036
Initial 63 features	178	0	0	0	0
u023	0	149	0	0	0
u024	1	1	129	42	18
u033	0	200	0	0	0
u036	0	0	1	0	102

Initial RF confusion matrix above is shown row-wise in the same class order: u018, u023, u024, u033, u036.

True \ Pred.	u018	u023	u024	u033	u036
u018	178	0	0	0	0
u023	0	149	0	0	0
u024	0	0	179	5	7
u033	0	200	0	0	0
u036	0	0	0	0	103

8.3 SVM

SVM with an RBF kernel was trained using the same 90 handcrafted features. Feature scaling was important because the feature magnitudes differed dramatically, for example spectral-energy features were hundreds of thousands while zero-crossing-rate values were fractions.

Baseline SVM result

Earlier properly scaled final-test accuracy: 74.79%. The corresponding confusion matrix showed u033 as a difficult class and strong performance on u018, u023 and u036.

Metric	Value
Accuracy	74.79%
Precision/Recall/F1 pattern	High for u023 and u036; u033 remained problematic
Main lesson	SVM is highly sensitive to feature scaling for this feature set

True \ Pred.	u018	u023	u024	u033	u036
u018	178	0	0	0	0

True \ Pred.	u018	u023	u024	u033	u036
u023	0	149	0	0	0
u024	0	0	184	4	3
u033	0	10	190	0	0
u036	0	0	0	0	103

8.4 1D CNN

The 1D CNN used raw 300 x 9 sensor windows rather than the 90 handcrafted features. The architecture used two Conv1D blocks with ReLU and max pooling, followed by adaptive average pooling, flattening, and a linear classifier for 5 classes.

Layer	Configuration
Input	9 channels x 300 time samples
Conv1D block 1	9 -> 32 filters, kernel 5, ReLU, MaxPool1d(2)
Conv1D block 2	32 -> 64 filters, kernel 5, ReLU, MaxPool1d(2)
Classifier	AdaptiveAvgPool1d(1) -> Flatten -> Linear(64, 5)

An early development evaluation produced 75.93% test accuracy after repeatedly inspecting the test set during development. A later recording-level validation protocol was used to obtain the proper final CNN test result of 74.69%.

8.5 1D CNN results

Version	Result
Development result	75.93%
Proper validation/test protocol	Best validation 98.97% at epoch 12; final test 74.69%

True \ Pred.	u018	u023	u024	u033	u036
u018	142	0	0	0	0
u023	0	182	0	0	0
u024	0	0	176	0	14
u033	93	0	97	0	0
u036	0	0	0	0	102

The final CNN confusion matrix shows a major generalization problem for u033: none of the 190 u033 test windows were classified as u033; they were distributed mainly into u018 and u024.

8.6 CNN-LSTM

A CNN-LSTM was tested as an advanced architecture intended to combine local feature extraction from CNN layers with temporal sequence learning from an LSTM.

Result	Value
Best validation accuracy	98.71%
Final test accuracy	53.35%
Main observation	Severe validation-to-test generalization gap

True \ Pred.	u018	u023	u024	u033	u036
u018	142	0	0	0	0
u023	0	174	0	8	0
u024	9	0	12	168	1
u033	139	0	51	0	0
u036	0	0	0	0	102

The large validation-to-test gap indicates strong overfitting or session-specific learning in this sample experiment. The result was retained as a valid negative finding rather than repeatedly tuning the model against the same test set.

9. Recording-Level Train / Validation / Test Design

For the later deep-learning experiment, a clean recording-level split was established to avoid leakage from overlapping windows of the same recording.

Set	Recording count / window count	Purpose
Training	31 recordings / 5056 windows in the later CNN setup	Model training
Validation	5 recordings / 778 windows	Model/epoch selection
Final Test	5 recordings / 806 windows	Final held-out evaluation

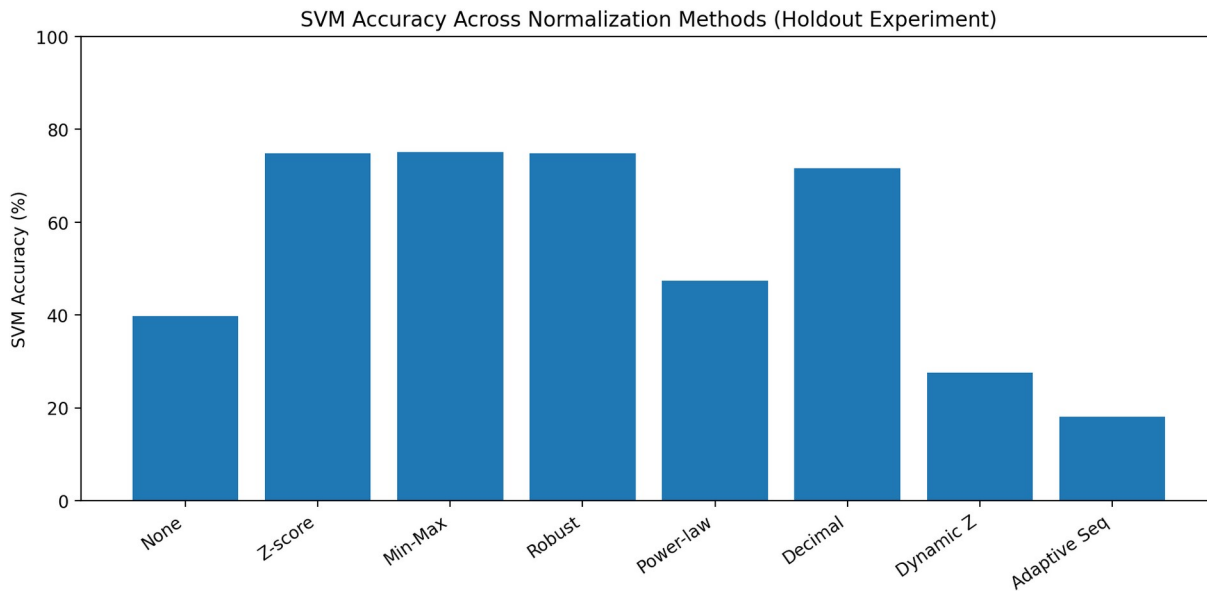
The five test recordings were u023_w012, u018_w014, u024_w008, u033_w007, and u036_w005. Train, validation, and test recording intersections were all empty. All five selected users appeared in all three sets.

10. Normalization Experiments

A major Phase 1 goal was to quantify how preprocessing changes model performance. The first controlled normalization comparison used SVM because SVM is particularly sensitive to feature scale.

Normalization	SVM Accuracy (%)	Interpretation
None	39.71	Strong baseline; large feature-scale differences harm SVM performance.
Z-score	74.79	Large improvement over unscaled features.
Linear (Min-Max)	75.15	Highest result in this holdout normalization comparison.
Robust (extra)	74.79	Comparable to Z-score; extra comparison, not one of the guide's eight named methods.
Power-law (gamma=0.5, signed)	47.38	Much lower than the linear and Z-score approaches under this implementation.
Decimal-place	71.62	Improved strongly over no normalization but below Z-score/Min-Max.
Dynamic Z-score (row-wise)	27.53	Very poor; row-wise standardization likely removes useful absolute information.
Adaptive Sequential (running statistics)	18.15	Very poor under the operational definition used.

These results are specific to the chosen SVM, feature representation, recording split, and operational definitions. They should not be presented as universal rankings of normalization methods.



11. Four Validation Methods

The guide required four validation strategies. The experiments below were carried out at the recording level to reduce leakage from overlapping windows.

11.1 Holdout validation

A 20% recording-level holdout was created from the 36 recordings in the SVM training pool. The final test set remained untouched. Results: Z-score 86.31% accuracy, FAR 3.44%, FRR 23.15%; Linear/Min-Max 85.40% accuracy, FAR 3.62%, FRR 23.50%.

11.2 5-Fold Cross-Validation

Five folds were created at the recording level. The scaler was fitted inside each training fold and then applied to that fold's validation recordings.

Normalization	Mean fold accuracy	Fold std	Pooled accuracy	FAR	FRR
Z-score	85.81%	9.33%	87.12%	3.44%	22.53%
Linear (Min-Max)	84.24%	6.50%	84.76%	4.15%	24.98%

11.3 Stratified 5-Fold Cross-Validation

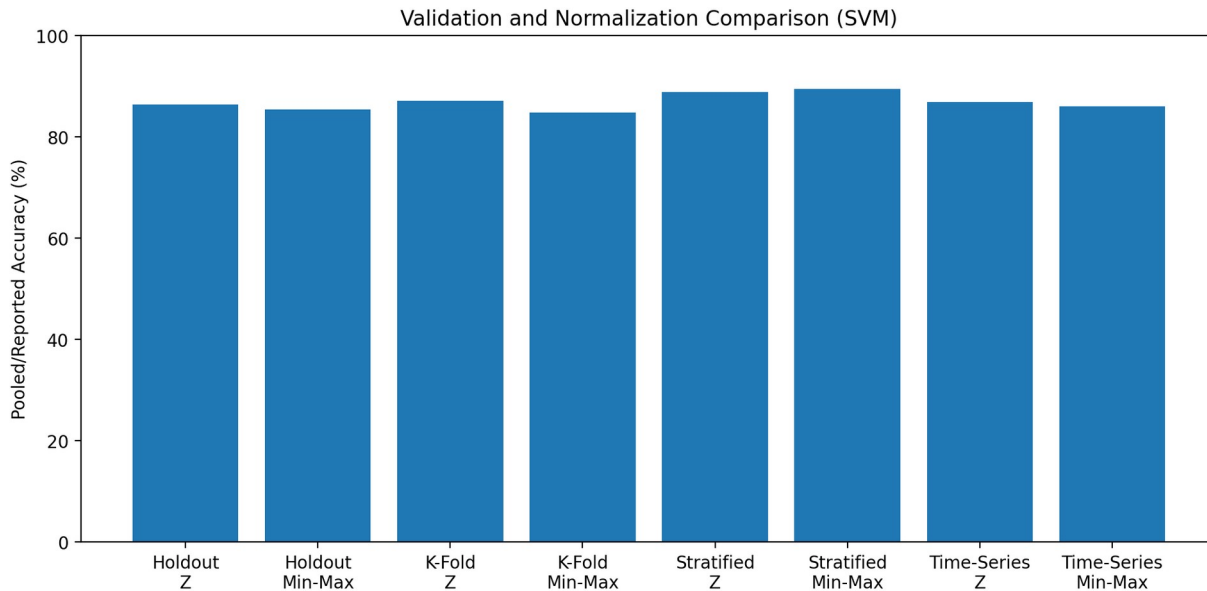
Stratification was applied at the recording level so that class representation was preserved as far as possible. A scikit-learn warning occurred because the least-populated class had only four recordings, fewer than the requested five folds. The experiment was retained with this limitation explicitly documented.

Normalization	Mean fold accuracy	Fold std	Pooled accuracy	FAR	FRR
Z-score	89.15%	5.14%	88.80%	2.97%	20.52%
Linear (Min-Max)	89.88%	5.21%	89.45%	2.73%	19.34%

11.4 Time-Series Split

Time-Series Split preserved the order of the recording/session indices used for the sample experiment. Because different subjects have their own w-number sequences, the ordering should be described as ordered recording/session index rather than as one universal clock.

Normalization	Mean fold accuracy	Fold std	Pooled accuracy	FAR	FRR
Z-score	85.70%	19.18%	86.84%	3.17%	24.77%
Linear (Min-Max)	84.72%	18.30%	85.98%	3.78%	24.14%



12. Evaluation Metrics and Confusion Matrix

Metric	Meaning used in project
Accuracy	Correct predictions divided by total predictions.
Precision	Among samples predicted as a class, the fraction that truly belongs to that class.
Recall	Among all true samples of a class, the fraction correctly identified.
F1-score	Harmonic balance of precision and recall.
FAR	For the current multiclass experiment, macro-averaged one-vs-rest false-acceptance rate: $FP/(FP+TN)$.
FRR	For the current multiclass experiment, macro-averaged one-vs-rest false-rejection rate: $FN/(TP+FN)$.
Confusion matrix	Rows = true classes, columns = predicted classes; diagonal entries are correct predictions.

FAR/FRR are naturally defined for binary accept/reject settings. Because the current IDNet experiment is five-class user identification, a one-vs-rest macro interpretation was used consistently for the reported FAR/FRR values. This convention should be confirmed against the guide's preferred definition before final submission.

13. Master Phase 1 Results Table (Completed Experiments)

Validation	Normalization	Model	Accuracy (%)	FAR (%)	FRR (%)
Holdout	Z-score	SVM	86.31	3.44	23.15
Holdout	Linear (Min-Max)	SVM	85.40	3.62	23.50
5-Fold CV	Z-score	SVM	87.12	3.44	22.53
5-Fold CV	Linear (Min-Max)	SVM	84.76	4.15	24.98
Stratified 5-Fold	Z-score	SVM	88.80	2.97	20.52
Stratified 5-Fold	Linear (Min-Max)	SVM	89.45	2.73	19.34
Time-Series Split	Z-score	SVM	86.84	3.17	24.77
Time-Series Split	Linear (Min-Max)	SVM	85.98	3.78	24.14

Best completed combination

Stratified 5-Fold + Linear (Min-Max) normalization + SVM: pooled accuracy 89.45%, FAR 2.73%, FRR 19.34%. This is the best combination among the completed normalization/validation experiments currently recorded.

Lowest completed accuracy in the master table

5-Fold Cross-Validation + Linear (Min-Max) + SVM: 84.76% pooled accuracy.

The earlier fold-level means were also recorded: Stratified 5-Fold + Linear = 89.88% mean fold accuracy (5.21% fold standard deviation), and Z-score = 89.15% (5.14%). The master table uses pooled predictions for Accuracy/FAR/FRR so that these metrics share a common aggregation convention.

14. Model Integration / Ensemble Experiment

The guide also suggested integrating models. A simple majority-voting ensemble was tested using Random Forest, SVM, and 1D CNN on the exact same common holdout windows. The earlier attempt exposed a mismatch of 42 windows between the pre-existing feature table and the raw-window CNN pipeline. Instead of trimming or fabricating rows, the 90-feature set was regenerated from the exact same 300 x 9 windows used by the CNN.

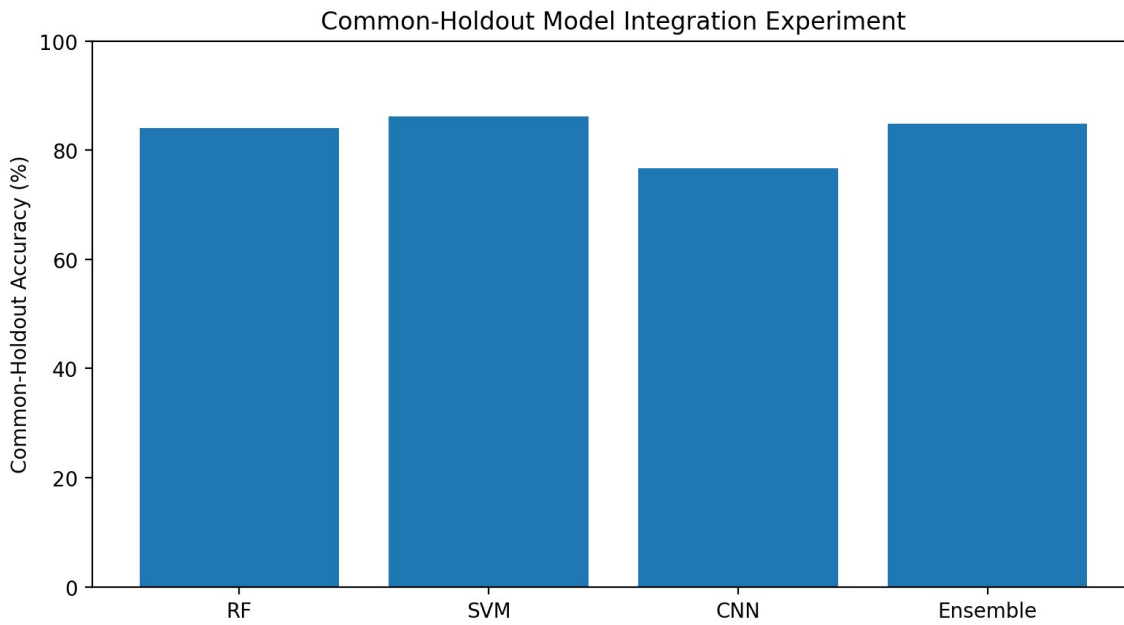
14.1 Alignment correction

Check	Result
CNN training windows	4696
Generated 90-feature training rows	4696
CNN validation windows	1138
Generated 90-feature validation rows	1138
Recording/window order matched	True
Label order matched	True

14.2 Ensemble results

Model	Accuracy (%)
Random Forest	84.01
SVM + Z-score	86.20
1D CNN + Z-score	76.63
RF + SVM + CNN majority vote	84.80

Ensemble metric	Value
Accuracy	84.80%
FAR	3.80%
FRR	25.03%

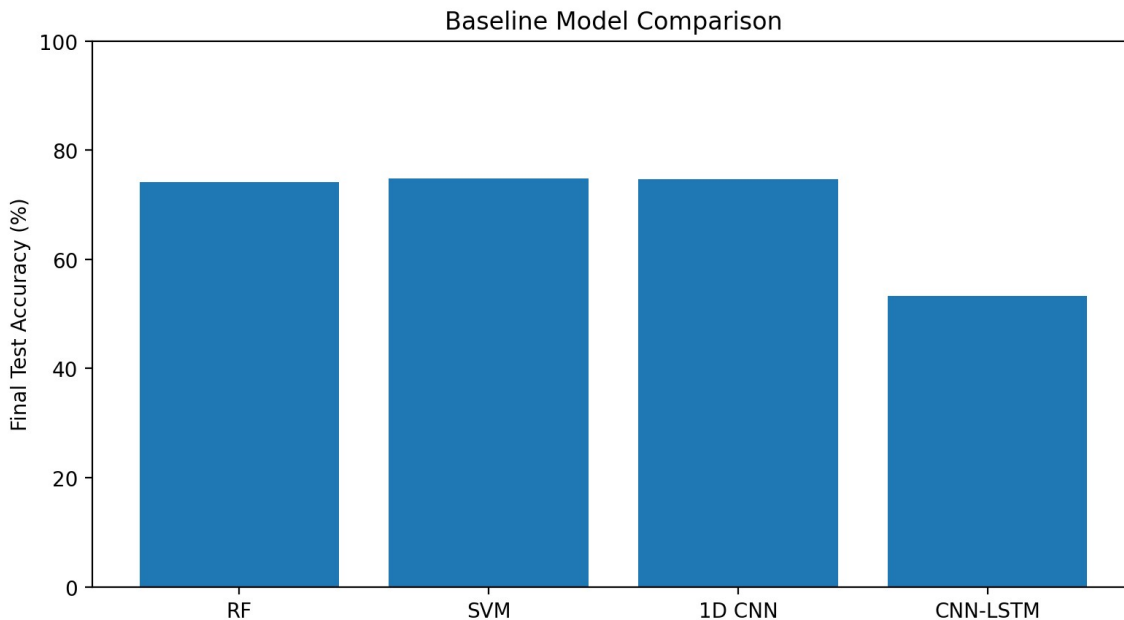


Interpretation: the simple majority-voting ensemble did not outperform the SVM on this common holdout split (84.80% versus 86.20%). This is a valid experimental finding. It indicates that the three models did not provide enough complementary errors for simple voting to improve the result.

14.3 Ensemble confusion matrix

True \ Pred.	u018	u023	u024	u033	u036
u018	380	0	0	0	0
u023	0	279	1	0	0
u024	2	2	210	45	0
u033	2	0	4	96	0
u036	0	3	114	0	0

15. Overall Baseline Model Comparison



Model	Input	Final test accuracy
Random Forest	90 handcrafted features	74.18%
SVM	90 handcrafted features	74.79%
1D CNN	Raw 300 x 9 sensor windows	74.69%
CNN-LSTM	Raw 300 x 9 sensor windows	53.35%

The three stronger individual baselines were very close in final held-out test accuracy. The CNN-LSTM was substantially lower, despite a validation accuracy of 98.71%, demonstrating that more complex architecture does not guarantee better generalization.

16. Generalization and Session-Variability Investigation

A major observation was the gap between high validation accuracy and lower final test performance. This prompted a recording-level inspection, especially for u033.

Recording	Role	Key observations
u033_w001-w004	Training	Earlier u033 recordings used for learning.
u033_w005	Validation	260.22 s; technically valid.
u033_w006	Review required	478.70 s; timestamp-order issues across all three main sensors.
u033_w007	Test	479.88 s; technically valid; test performance showed strong confusion with u018/u024 in the CNN experiment.

Feature-level descriptive statistics for u033 recordings were inspected. Differences in accelerometer and magnetometer statistics between sessions supported the hypothesis that recording/session variability is a major challenge in the sample experiment.

17. Important Technical Errors and How They Were Resolved

Problem encountered	Cause	Resolution
Variable-length raw windows	Different effective sampling/window lengths	Moved to fixed 60 Hz master timeline; every full window = 300 x 9.
X_train_raw was inhomogeneous	Windows had different lengths	Rebuilt windows with fixed length.
X_test_scaled / X_train_scaled not defined	Fresh Jupyter kernel	Re-ran the required preprocessing/scaling cells.
PyTorch missing	Torch not installed in environment at that point	Installed/used the correct environment, then rebuilt tensors.
Many NameErrors after restart	Notebook code remained saved, but kernel memory was cleared	Restored functions/variables in dependency order.
SVM rejected u023_w001 as float	Metadata column “recording” was included as a feature	Selected only 90 numeric feature columns, excluding recording and label.
Decimal-place normalization broadcasting error	90 scaling factors were aligned incorrectly by pandas	Converted scaling vector to NumPy and applied column-wise.
CNN/SVM ensemble had 1138 vs 1096 samples	Different earlier window/feature generation paths	Regenerated 90 features from the exact CNN windows.
Missing plotting imports	ConfusionMatrixDisplay / matplotlib not in memory	Restored imports and reran only plotting cells.

18. Files and Result Artifacts Generated

The project results folder was used as the persistent record. Important outputs included model files, CSV result tables, confusion matrices, plots, and textual observations.

Artifact type	Examples / role
Models	Random Forest .pkl, SVM .pkl, CNN .pth, CNN-LSTM .pth, ensemble CNN checkpoint
Model comparison CSVs	Model-wise accuracy tables and ensemble results
Normalization CSVs	Normalization holdout results
Validation CSVs	Holdout, K-Fold, Stratified K-Fold, Time-Series metrics
Confusion matrices	Best, worst, and ensemble matrices in PNG/CSV form
Quality records	Recording quality and review-required information
Documentation	Progress diary / technical documentation

19. Daily Project Diary Entries (Suggested Record Format)

The following entries can be copied into the project diary and expanded with the actual date whenever the team records the work officially.

Day 1 - Dataset inspection: Inspected the IDNet sample dataset structure, identified recording folders, sensor files, user IDs, and session naming.

Day 2 - Data quality: Scanned recordings for missing values, duplicates, timestamp ordering, durations, sampling rates, constant signals, and status classification.

Day 3 - Sensor analysis: Compared accelerometer, gyroscope, and magnetometer sampling rates and identified the need for synchronization.

Day 4 - Synchronization: Implemented common interval selection, fixed 60 Hz master timeline, and interpolation for the three sensor streams.

Day 5 - Windowing: Implemented 5-second windows with 50% overlap. Corrected variable-length windows by fixing the synchronized sampling rate at 60 Hz.

Day 6 - Handcrafted features: Implemented 90 features: time-domain descriptors, zero-crossing rate, dominant frequency, and spectral energy.

Day 7 - Random Forest: Trained initial and improved Random Forest models; compared confusion matrices and feature importance.

Day 8 - SVM: Trained RBF SVM on 90 features and observed the strong effect of feature scaling.

Day 9 - 1D CNN: Trained a CNN directly on 300 x 9 raw windows; inspected training curves and confusion matrices.

Day 10 - CNN-LSTM: Tested CNN-LSTM; observed a large validation-to-test gap and retained the finding as an overfitting/generalization observation.

Day 11 - Proper split: Created recording-level train/validation/test splits and verified no recording overlap between sets.

Day 12 - Data-quality investigation: Investigated u033 session variability and timestamp anomalies in u033_w006.

Day 13 - Normalization: Compared unscaled, Z-score, Linear/Min-Max, Robust, Power-law, Decimal-place, Dynamic Z-score, and Adaptive Sequential preprocessing under the SVM holdout experiment.

Day 14 - Holdout validation: Created recording-level holdout validation and calculated Accuracy, FAR, and FRR.

Day 15 - K-Fold: Implemented recording-level 5-Fold Cross-Validation with scaling fitted inside each fold.

Day 16 - Stratified K-Fold: Implemented recording-level stratified 5-Fold; documented the small-class warning and calculated pooled Accuracy/FAR/FRR.

Day 17 - Time-Series Split: Implemented ordered recording/session split and calculated Accuracy/FAR/FRR.

Day 18 - Best/worst analysis: Created master result table and confusion matrices for best and worst completed combinations.

Day 19 - Model integration: Aligned CNN windows and 90-feature rows exactly, retrained RF/SVM on the common holdout data, and evaluated RF + SVM + CNN majority voting.

Current - Report preparation: Consolidated procedures, metrics, results, lessons learned, limitations, and remaining guide-specific items into the Phase 1 project diary.

20. Simple Explanation of the Main Concepts

Concept	Simple meaning in this project
Sensor synchronization	Put accelerometer, gyroscope, and magnetometer readings onto one common time grid.
Windowing	Cut continuous motion data into fixed 5-second pieces so each sample has the same shape.
Feature extraction	Convert each window into 90 summary numbers for traditional ML.
Normalization	Rescale the feature values so differences in magnitude do not dominate the model.
Random Forest	Many decision trees combined into one classifier for tabular features.
SVM	Finds separating boundaries between classes; sensitive to feature scaling.
1D CNN	Learns local temporal patterns directly from the raw sensor sequence.
CNN-LSTM	CNN learns local patterns; LSTM attempts to model longer temporal relationships.
Holdout	Keep a fixed subset aside for validation.
K-Fold	Rotate different subsets as validation folds and average performance.
Stratified K-Fold	K-Fold while trying to preserve class proportions in each fold.
Time-Series Split	Train on earlier ordered data and validate on later ordered data.
Confusion matrix	Shows exactly which classes are being predicted correctly or confused.
FAR	False positive rate under the current one-vs-rest macro convention.
FRR	False negative rate under the current one-vs-rest macro convention.

21. Limitations and Remaining Work

- IDNet was used only as a sample dataset for pipeline validation. The sample labels are user IDs, not Parkinson disease labels.
- The current normalization comparison is not yet the full guide-specified 8 x 4 matrix because the exact intended procedures for Standard-deviation normalization and Canonical Multi-Vendor Schema Mapping were not available during the experiment.
- Dynamic Z-score, Power-law, and Adaptive Sequential methods were implemented using explicit operational definitions chosen for the experiment. These definitions should be confirmed with the guide before formal submission.
- FAR/FRR are currently computed as macro-averaged one-vs-rest measures for the five-class sample problem. The guide should confirm whether this is the desired convention.
- The stratified 5-fold recording-level split had a class-size limitation: one class had only four recordings.
- The time-series split used ordered session indices (w-number) rather than a single absolute time axis across all users.
- The simple majority-voting ensemble did not improve over SVM on the common holdout split.

22. Current Phase 1 Conclusion

The Phase 1 sample-data study successfully demonstrated an end-to-end sensor classification pipeline. The data were inspected, quality-checked, synchronized, segmented into fixed 300 x 9 windows, and represented both as raw sequences and 90 handcrafted features. Random Forest, SVM, 1D CNN, and CNN-LSTM models were trained and

evaluated. Multiple normalization methods and four validation strategies were then tested for SVM, with Accuracy, FAR, FRR, and confusion matrices recorded.

Among the completed normalization-validation combinations, the strongest recorded combination was Stratified 5-Fold Cross-Validation with Linear (Min-Max) normalization and SVM, giving 89.45% pooled accuracy, 2.73% FAR, and 19.34% FRR. A separate common-holdout model-integration experiment using Random Forest, SVM, and 1D CNN majority voting produced 84.80% accuracy, 3.80% FAR, and 25.03% FRR, which did not exceed the SVM result on that same holdout.

Transition to the next phase

The validated pipeline is ready to be adapted to the actual Parkinson-labelled dataset. The next scientific objective is to replace user-identification labels with the correct clinical classes and preserve the same leakage-controlled preprocessing and evaluation discipline.

Appendix A - Key Numerical Results

Experiment	Result
RF initial 63-feature test	67.97%
RF improved 90-feature test	74.18%
SVM 90-feature final test	74.79%
1D CNN development test	75.93%
1D CNN proper final test	74.69%
CNN-LSTM best validation	98.71%
CNN-LSTM final test	53.35%
SVM no normalization	39.71%
SVM + Z-score holdout normalization experiment	74.79%
SVM + Linear/Min-Max holdout normalization experiment	75.15%
SVM + Robust holdout normalization experiment	74.79%
SVM + Power-law holdout normalization experiment	47.38%
SVM + Decimal-place holdout normalization experiment	71.62%
SVM + Dynamic Z-score holdout normalization experiment	27.53%
SVM + Adaptive Sequential holdout normalization experiment	18.15%
Best completed normalization+validation combination	Stratified 5-Fold + Linear/Min-Max + SVM
Best completed pooled accuracy	89.45%
Best completed FAR	2.73%
Best completed FRR	19.34%
Common-holdout RF+SVM+CNN ensemble	84.80%

Appendix B - Reproducibility and Data-Saving Practice

The project used a local project directory and a results folder to persist models and experiment outputs. Notebook code was kept in the Jupyter notebook, while model and result artifacts were saved to disk. A full project-folder backup was also recommended before closing the environment.

- Save the notebook with Ctrl+S before shutting down Jupyter.
- Preserve the results folder and processed-data folder.
- Keep a complete backup copy of the project folder.
- When reopening the notebook, remember that variables in kernel memory are cleared, but saved files remain.

Appendix C - Team Review / Guide Comments

Item	Notes / Guide comments
Normalization definitions	
FAR/FRR convention	
Required model set for integration	
Canonical Multi-Vendor Schema Mapping procedure	
Standard-deviation normalization procedure	
Other corrections / remarks	